

LIVERPOOL JOHN LENNON, United Kingdom

WMO: 033233

Lat: 53.334N

Lon: 2.850W

Elev: 80

StdP: 14.65

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.3	30.3	21.6	16.2	32.7	24.5	18.6	34.2	37.0	47.1	32.6	47.2	6.4	100	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	77.2	64.2	73.6	62.6	71.2	61.9	65.7	73.7	64.2	71.4	62.8	68.9	9.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.8	85.7	68.7	61.1	80.6	67.2	60.4	78.6	66.6	30.6	73.7	29.4	71.7	28.4	69.2	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.5	24.8	21.9	DB	23.0	83.2	4.0	3.7	20.1	85.9	17.8	88.1	15.5	90.2	12.6	92.9	(4)
			WB	22.2	69.4	4.0	2.4	19.3	71.1	17.0	72.5	14.7	73.9	11.8	75.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	41.6	42.1	44.9	48.8	54.1	59.1	62.6	62.2	58.2	52.7	46.5	42.3	(6)
		DBStd	8.99	5.34	5.18	4.82	4.90	4.91	4.33	4.21	3.69	3.79	4.63	5.11	6.12	(7)
		HDD50	1141	262	224	169	77	14	0	0	0	26	125	244		(8)
		HDD65	5053	725	641	624	485	338	185	102	105	208	382	555	703	(9)
		CDD50	1619	2	3	10	42	142	273	390	377	245	109	20	6	(10)
		CDD65	56	0	0	0	0	1	8	27	17	3	0	0	0	(11)
		CDH74	297	0	0	0	0	8	56	153	73	6	1	0	0	(12)
		CDH80	46	0	0	0	0	0	7	26	13	0	0	0	0	(13)
(14)	Wind	WSAvg	10.5	11.8	11.7	11.4	10.5	10.3	9.8	9.6	9.7	10.6	10.6	10.9		(14)
(15)	Precipitation	PrecAvg	31.5	2.8	2.1	1.9	2.1	2.3	2.5	2.6	2.9	2.7	3.3	3.2	3.2	(15)
(16)		PrecMax	42.8	5.2	3.9	4.1	4.4	4.3	7.0	5.2	6.7	5.2	6.7	6.9	5.9	(16)
(17)		PrecMin	22.8	0.4	0.3	0.4	0.5	0.4	0.7	0.7	0.4	0.7	1.6	1.1		(17)
(18)		PrecStd	4.7	1.3	0.9	0.9	1.1	1.0	1.5	1.1	1.2	1.2	1.4	1.2	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	55.7	57.0	60.7	68.4	75.1	80.4	83.8	81.0	74.9	66.3	60.5	57.3		(19)
(20)		0.4% MCWB	51.6	52.2	52.5	56.8	62.0	64.1	66.1	66.2	63.7	60.6	56.8	53.6		(20)
(21)		2% DB	53.5	53.8	57.1	62.9	70.2	73.8	78.4	75.0	70.0	62.9	57.4	55.2		(21)
(22)		2% MCWB	49.6	49.6	51.0	54.2	59.7	62.4	64.1	64.0	61.8	58.6	54.2	52.0		(22)
(23)		5% DB	51.5	51.8	54.0	59.3	66.2	70.0	73.8	71.5	66.6	61.1	55.6	53.3		(23)
(24)		5% MCWB	48.0	47.8	49.0	52.0	57.1	60.6	62.8	62.9	59.7	56.9	52.1	50.2		(24)
(25)		10% DB	49.7	50.0	52.1	57.0	62.6	67.6	70.2	68.3	64.6	59.3	53.7	50.4		(25)
(26)		10% MCWB	46.9	46.7	47.5	50.7	55.4	59.0	61.8	61.2	58.2	55.3	50.5	47.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	52.5	52.7	53.9	57.6	63.6	66.9	68.2	67.4	65.3	61.6	57.2	54.5		(27)
(28)		0.4% MCDB	55.4	56.3	58.7	65.8	73.0	75.5	77.7	75.4	71.7	64.9	59.8	56.6		(28)
(29)		2% WB	50.3	50.5	51.8	55.0	60.4	63.6	65.8	65.4	63.1	59.4	55.1	52.4		(29)
(30)		2% MCDB	52.6	53.3	56.0	61.5	68.3	71.1	74.9	72.4	68.3	62.4	57.5	55.0		(30)
(31)		5% WB	48.5	48.6	50.2	52.8	58.3	61.5	64.1	63.9	60.9	57.6	52.9	50.6		(31)
(32)		5% MCDB	50.7	51.2	53.8	58.7	64.9	68.3	72.0	70.0	65.3	60.8	55.5	52.8		(32)
(33)		10% WB	46.8	46.9	48.5	51.1	56.0	59.8	62.5	62.5	59.4	55.8	51.3	48.4		(33)
(34)		10% MCDB	49.1	49.6	52.1	56.1	61.8	66.0	69.0	67.4	63.6	59.1	53.7	50.4		(34)
(35)	Mean Daily Temperature Range	MDBR	7.9	9.1	10.7	12.3	13.1	12.7	12.5	11.6	11.4	10.0	8.3	8.1		(35)
(36)		5% DB MCDBR	9.8	10.4	13.1	16.9	19.4	18.9	19.2	16.5	15.0	10.7	9.2	9.5		(36)
(37)		5% DB MCWBR	9.0	8.4	9.1	10.2	11.2	10.0	9.0	8.6	8.8	8.1	8.1	8.7		(37)
(38)		5% WB MCDBR	9.3	10.1	12.1	15.2	17.3	16.2	16.5	14.0	12.8	10.3	8.9	9.4		(38)
(39)		5% WB MCWBR	8.9	8.7	9.1	9.9	10.7	9.8	8.9	8.2	8.8	8.5	8.2	8.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.341	0.371	0.394	0.398	0.395	0.395	0.381	0.368	0.350	0.326	0.316		(40)
(41)		taud	2.321	2.375	2.300	2.252	2.285	2.323	2.328	2.383	2.420	2.432	2.385	2.297		(41)
(42)		Ebn at Noon	202	240	256	265	270	272	269	267	255	234	202	180		(42)
(43)		Edh at Noon	19	25	32	38	39	38	37	33	29	23	18	16		(43)
(44)	All-Sky Solar Radiation	RadAvg	204	419	785	1214	1541	1616	1527	1254	900	500	248	154		(44)
(45)		RadStd	20	45	67	112	109	151	151	82	67	43	26	15		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	+1.08	+1.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page